

HEKKINGEN FYR, Norway

WMO: 010150

Lat: 69.601N

Lon: 17.831E

Elev: 14

StdP: 101.16

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.0	-7.8	-15.0	1.0	-8.1	-13.4	1.2	-6.8	22.7	4.5	20.4	1.6	10.0	160	0.853

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.9	18.9	14.4	17.4	13.4	16.0	12.7	14.9	18.0	13.9	16.7	13.0	15.5	5.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.3	9.5	16.5	12.3	8.9	15.4	11.5	8.5	14.5	41.8	18.2	39.0	16.9	36.7	15.6	19.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	15.6	13.5	DB	-10.7	22.1	2.5	2.5	-12.5	23.9	-14.0	25.4	-15.4	26.8	-17.2	28.6	(4)
(5)			WB	-11.7	16.4	2.4	1.5	-13.4	17.5	-14.8	18.3	-16.1	19.2	-17.8	20.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-1.2	-1.2	-0.3	2.9	6.5	9.0	12.1	12.0	9.4	5.4	2.3	0.7	
	DBStd	5.66	3.72	3.75	3.12	2.88	2.98	2.70	2.64	2.37	2.75	3.09	3.11	3.38	
	HDD10.0	2076	347	313	318	213	115	49	7	7	43	145	232	287	
	HDD18.3	4923	605	547	577	463	367	279	194	197	267	401	482	546	
	CDD10.0	196	0	0	0	0	7	21	72	68	26	3	0	0	
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0	
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	6.6	8.9	8.5	7.6	6.1	5.5	4.7	4.2	4.6	6.1	6.8	7.8	8.5	
Precipitation	PrecAvg	1043	94	85	92	64	52	55	68	77	110	134	105	104	
	PrecMax	1286	229	162	217	161	140	113	136	188	227	210	229	182	
	PrecMin	802	0	15	20	0	4	13	15	27	27	24	9	0	
	PrecStd	120	63	34	57	34	31	23	36	37	55	50	57	51	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	8.2	8.5	11.8	17.1	20.2	21.6	19.9	17.2	13.9	10.4	9.3	
		MCWB	5.3	5.4	4.5	7.0	10.8	13.0	15.8	15.4	12.3	9.8	7.8	6.3	
	2%	DB	6.5	6.7	7.0	9.7	14.2	16.7	19.3	18.2	15.4	12.1	8.9	8.0	
		MCWB	4.1	4.0	3.9	6.1	9.6	12.0	14.8	14.4	11.7	9.0	6.8	5.5	
	5%	DB	5.3	5.6	5.6	8.1	12.3	14.7	17.8	16.8	14.3	10.8	7.7	6.5	
		MCWB	3.2	3.3	2.9	5.1	8.2	11.2	13.9	13.5	11.1	8.2	5.3	4.1	
	10%	DB	3.9	4.0	4.2	6.9	10.7	13.0	16.3	15.5	13.2	9.7	6.5	5.2	
		MCWB	2.0	2.1	2.0	4.3	7.3	10.3	13.1	12.8	10.5	7.4	4.4	3.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	5.9	5.2	7.4	11.5	13.9	16.4	16.6	13.3	10.2	8.2	6.9
			MCDB	7.5	7.8	7.6	10.9	16.1	19.1	20.4	18.5	16.1	12.8	9.8	8.9
2%		WB	4.4	4.2	4.1	6.3	9.7	12.5	15.2	14.9	12.3	9.3	7.1	5.6	
		MCDB	6.3	6.3	6.4	9.0	13.7	16.1	18.7	17.4	14.6	11.7	8.9	7.7	
5%		WB	3.2	3.3	3.2	5.4	8.6	11.4	14.2	13.9	11.5	8.4	5.5	4.3	
		MCDB	5.1	5.3	5.4	7.9	11.8	14.4	17.4	16.2	13.7	10.5	7.4	6.3	
10%		WB	2.0	2.2	2.1	4.5	7.6	10.4	13.2	13.0	10.8	7.5	4.5	3.1	
		MCDB	3.8	4.0	4.0	6.7	10.2	12.8	15.9	15.3	12.9	9.5	6.4	5.0	
Mean Daily Temperature Range		5% DB	MDBR	3.2	3.2	3.2	3.8	4.1	3.6	3.9	3.7	3.2	2.8	2.9	3.2
			MCDBR	4.6	4.9	4.4	5.2	6.6	6.3	6.9	5.8	4.7	4.2	4.2	4.9
	MCWBR		3.9	3.9	3.5	3.4	3.7	3.7	3.9	3.4	2.8	3.1	3.4	4.0	
	5% WB	MCDBR	4.6	4.9	4.2	5.0	6.2	5.7	6.3	5.2	4.3	4.0	4.0	4.8	
		MCWBR	3.9	4.0	3.6	3.4	3.8	3.7	3.9	3.4	2.9	3.1	3.5	4.1	
Clear-Sky Solar Irradiance	taub	0.374	0.273	0.318	0.334	0.343	0.342	0.346	0.335	0.319	0.278	N/A	N/A		
	taud	2.586	2.101	2.193	2.320	2.438	2.496	2.516	2.527	2.528	2.219	N/A	N/A		
	Ebn at Noon	44	449	678	802	844	856	838	805	713	442	0	0		
	Edh at Noon	8	42	75	91	91	89	84	74	56	36	0	0		
All-Sky Solar Radiation	RadAvg	0.01	0.26	1.26	3.02	4.53	4.90	4.38	3.10	1.63	0.50	0.00	0.00		
	RadStd	0.00	0.05	0.18	0.31	0.32	0.64	0.44	0.26	0.17	0.08	0.00	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% DP	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page